THE MASTERMIND COACHING CLASSES
CBSE / RBSE • NURSERY – CLASS XII • JAIPUR
Road No. 5, VKIA, Jaipur — 9887258679

# ANSWER KEY & MARKING SCHEME SET 1

Chapter 12: **How Nature Works in Harmony** | Class 8 Science | Maximum Marks: 40

## SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(b)** producers | They make food. |
| 2 | **(d)** food chain | Sequence. |
| 3 | **(b)** deforestation | Removal of trees. |
| 4 | **(d)** decomposers | Break down dead matter. |
| 5 | **(d)** food web | Network. |
| 6 | **(a)** afforestation | Adds forests. |
| 7 | **(d)** herbivore | Plant-eater. |
| 8 | **(d)** producer | Plants. |
| 9 | **(c)** water pollution | Pollution. |
| 10 | **(a)** carnivore | Flesh-eater. |

## SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Living and non-living things interacting in an area (2). | 2 |
| 12 | Sequence of organisms in which one eats another (2). | 2 |
| 13 | Variety of living organisms (2). | 2 |
| 14 | Grass (1); Trees (1). | 1+1 |

## SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Biotic: fish, algae, frog (1); Abiotic: water, light, temperature (1); They interact (1). | 1+1+1 |
| 16 | Provide oxygen (1); Provide wood and food (1); Homes for animals (1). | 1+1+1 |
| 17 | Source of energy (1); Photosynthesis (1); Water cycle (1). | 1+1+1 |
| | **OR:** Producers make food (1); Consumers eat food (1); Decomposers recycle (1). | 1+1+1 |
| 18 | Bacteria (1); Fungi (1); Earthworms help in decay (1). | 1+1+1 |
| | **OR:** Reduce (1); Reuse (1); Recycle (1). | 1+1+1 |

## SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Biotic (1); Abiotic (1); Producers (1); Consumers (1); Decomposers (1). | 1+1+1+1+1 |
| | **OR:** Plant trees (1); 3 R's (1); Save water (1); Use public transport (1); Protect animals (1). | 1+1+1+1+1 |
| 20 | Definition (1); Grass-deer-tiger (1); Trophic levels (1); Energy flow (1); Decomposers (1). | 1+1+1+1+1 |
| | **OR:** Break dead matter (1); Return nutrients (1); Clean surroundings (1); Examples (1); Manure (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

## ANSWER KEY & MARKING SCHEME SET 2

Chapter 12: **How Nature Works in Harmony** | Class 8 Science | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(c)** top carnivore | Eaten by none. |
| 2 | **(c)** air pollution | Smoke and CO2. |
| 3 | **(a)** omnivores | Eat both. |
| 4 | **(a)** heat | Heat. |
| 5 | **(d)** wildlife | Conservation. |
| 6 | **(d)** abiotic | Non-living. |
| 7 | **(d)** 10 per cent | Rule. |
| 8 | **(b)** non-biodegradable | Does not decompose. |
| 9 | **(a)** abiotic | Non-living. |
| 10 | **(a)** carbon dioxide | For photosynthesis. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Natural home of an organism (2). | 2 |
| 12 | Smoke (1); Garbage (1). | 1+1 |
| 13 | Deer (1); Tiger (1). | 1+1 |
| 14 | Water (1); Air (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Grass-deer-tiger (1); Algae-fish-bird (1); Grain-mouse-snake (1). | 1+1+1 |
| 16 | Herbivores: cow, deer, goat (1); Carnivores: lion, tiger, eagle (1); Different feeding (1). | 1+1+1 |
| 17 | Jim Corbett (1); Ranthambore (1); Kaziranga (1). | 1+1+1 |
| | **OR:** Loss of habitat (1); Soil erosion (1); Less rain (1). | 1+1+1 |
| 18 | Evaporation (1); Condensation (1); Precipitation (1). | 1+1+1 |
| | **OR:** Prey increases (1); Plants overeaten (1); Balance lost (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Sun (1); Producers (1); Consumers (1); 10 per cent (1); Heat loss (1). | 1+1+1+1+1 |
| | **OR:** Grass (1); Grasshopper and mouse (1); Frog and snake (1); Hawk (1); Links (1). | 1+1+1+1+1 |
| 20 | Loss of habitat (1); Soil erosion (1); Floods (1); Climate change (1); Less oxygen (1). | 1+1+1+1+1 |
| | **OR:** Stable ecosystem (1); Food (1); Medicines (1); Beauty (1); Climate (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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# ANSWER KEY & MARKING SCHEME SET 3

Chapter 12: **How Nature Works in Harmony** | Class 8 Science | Maximum Marks: 40

## SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(c)** decomposers | Microbes. |
| 2 | **(c)** carnivores | Eat insects. |
| 3 | **(d)** oxygen | During photosynthesis. |
| 4 | **(b)** pulses | Nitrogen fixers. |
| 5 | **(a)** Sun | Sunlight. |
| 6 | **(b)** pollination | Transfer of pollen. |
| 7 | **(a)** clouds | Cooling. |
| 8 | **(d)** eater | Energy transfer. |
| 9 | **(b)** seed dispersal | Seeds in droppings. |
| 10 | **(c)** precipitation | Falling water. |

## SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Grass (1); Goat and tiger (1). | 1+1 |
| 12 | Bacteria (1); Fungi (1). | 1+1 |
| 13 | Cutting down of forests (2). | 2 |
| 14 | Plant trees (1); Avoid plastics (1). | 1+1 |

## SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Diseases (1); Death of fishes (1); Climate change (1). | 1+1+1 |
| 16 | Chains are short (1); Fewer top animals (1); More producers than consumers (1). | 1+1+1 |
| 17 | Network of food chains (1); Shows many feeding relations (1); More stable (1). | 1+1+1 |
| | **OR:** Use public transport (1); Plant trees (1); Dispose waste properly (1). | 1+1+1 |
| 18 | Bees and flowers (1); Birds and trees (1); Plants and animals gases (1). | 1+1+1 |
| | **OR:** Take in $CO_2$ (1); Give out oxygen (1); Trap dust (1). | 1+1+1 |

## SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Evaporation (1); Transpiration (1); Condensation (1); Rain (1); Run-off (1). | 1+1+1+1+1 |
| | **OR:** Bees (1); Birds (1); Bacteria (1); Lichens (1); Plants and animals (1). | 1+1+1+1+1 |
| 20 | Photosynthesis (1); Respiration (1); Decay (1); Burning (1); Balance (1). | 1+1+1+1+1 |
| | **OR:** Bacteria (1); Root nodules (1); Pulses (1); Nitrogen to plants (1); Soil fertility (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.